

energy later in the day. It may add or remove slots as it chooses. This may result in several iterations for the Energy Management System to reprocess updated forecasts until it has stabilized on an agreed set of changes.

If for any reason the ESA cannot accept the entire requested forecast adjustments then it SHALL reject the entire command.

If the Cause is not permitted, or the requested forecast exceeds any of the limits in the respective slots, then the command SHALL be rejected with CONSTRAINT_ERROR, otherwise if the ForecastID is valid, and the entire list of [SlotAdjustmentStruct](#) are accepted, the command status returned SHALL be SUCCESS, otherwise the command SHALL be rejected with FAILURE.

9.2.9.7. RequestConstraintBasedForecast Command

Allows a client to ask the ESA to recompute its Forecast based on power and time constraints.

ID	Name	Type	Constraint	Quality	Fallback	Conformance
0	Constraints	list[ConstraintsStruct]	max 10			M
1	Cause	Adjustment-CauseEnum	all			M

9.2.9.7.1. Constraints Field

This field SHALL indicate the series of turn up or turn down power requests that the ESA is being asked to constrain its operation within. These requests SHALL be in the future, SHALL be in chronological order, starting with the earliest start time, and SHALL NOT overlap in time.

For example, a grid event which requires devices to reduce power (turn down) between 4pm and 6pm and due to excess power on the grid overnight, may request ESAs to increase their power demand (turn up) between midnight and 6am.

If this ESA supports **PFR** this would have 2 entries in the list as follows:

Entry [0] - Turn Down		
StartTime	2023-10-23T16:00:00	
Duration	7200	(2 hours)
NominalPower	20	W
MaximumEnergy	40	Wh
Entry [1] - Turn Up		
StartTime	2023-10-24T00:00:00	
Duration	21,600	(6 hours)